

ITF23019: Parallel and Distributed Programming

Spring Semester, 2026

Lab12: Practice with Cloud Computing

Submission Deadline: April 22th, 2026 23:59

This lab helps to explain some Cloud Computing concepts, such as hardware virtualization, user interface, services, service provisioning etc...which are covered in the lectures. In this lab, you will perform some practice in an Azure Lab Virtual Machine. The first part of this document serves as a guide on how to connect to a virtual machine in the Microsoft Azure Cloud called **template machine**. This service is provided as a utility with **pay-as-you go** pricing model. Our university college covers the cost for your allocated time quota. To work efficiently, it is strongly recommended that you:

- Work on this lab assignment.
- Read the manual in this lab carefully before practice.
- Remember to shut down your virtual machine and sign out from your account when you have finished practicing.

Make sure you have received my invitation via your school email to access the cloud service. If you haven't received it, please contact me. In addition, please be aware that:

- You may experience some delays during your practice, so patience is important. Additionally, you may encounter new concepts such as **sandbox**, **HDFS**, **Ambari** etc., which you may not know before. However, it is sufficient to just use them for this lab without fully understanding what they actually are.
- The computing services provided for this lab may be more than what we actually need. However, the primary purpose of using this Cloud Computing service is for illustration and to give hands-on experience with using services from one of the three major service providers currently in the market.
- There may be several CC service providers that offer equal or even better services. Nonetheless, Microsoft Azure Cloud has been selected because HIOF uses Microsoft services, and this may be an opportunity to see the potential implications of being locked into a single service provider.

1 Connect to Azure Labs Virtual Machine

1.1 Start your Azure Virtual Machine

1. Click on the registration link you have received via your school email:

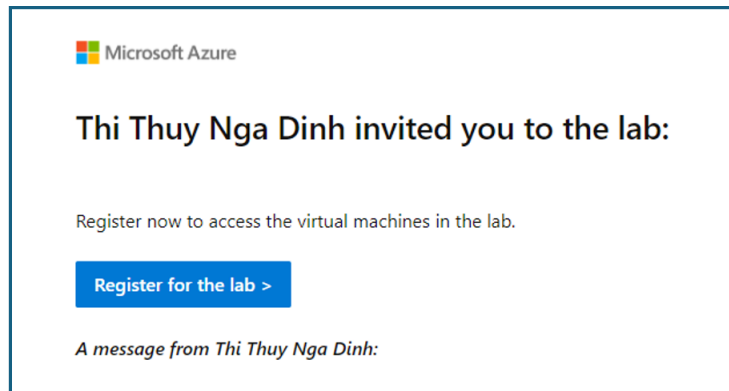


Figure 1: Click on 'Registration' link you get from your email.

2. You are now registered with the lab:

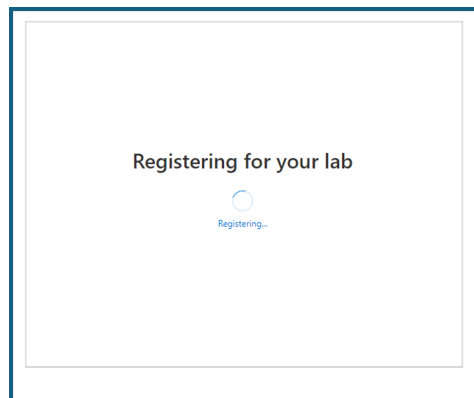


Figure 2: The first access to the Azure Labs.

3. You should see your Azure Labs Virtual Machine - Figure 3. It is currently in the **Stop** state. Pay attention to the **Start/Stop** switch which you will use in this lab.

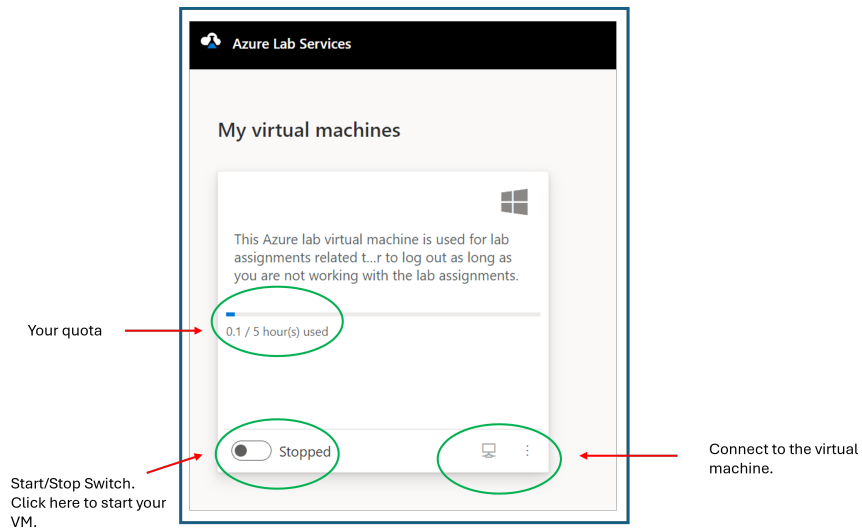


Figure 3: Your Azure Lab virtual machine.

4. Start your VM by clicking the **Start/Stop** switch. Your VM may remain in the **Starting** state for a while.

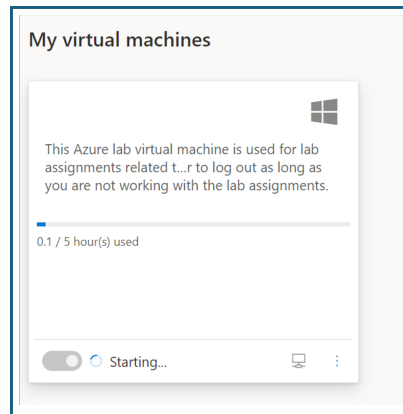


Figure 4: Your VM is in the **Starting** state....

5. After a while, your VM is ready in the **Running...** state.

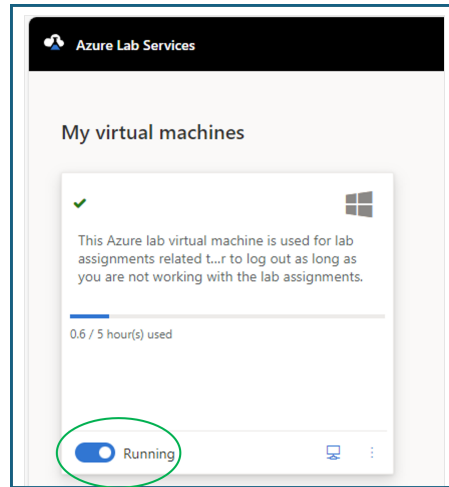


Figure 5: Your VM is in the **Running...** state.

Now, your VM is ready to connect to the template machine.

1.2 Connect to the Azure VM

Now that your VM is ready to connect to an Azure Virtual Machine. It is the template machine where you can have remote access to do your lab assignments. The template machine is look like this:

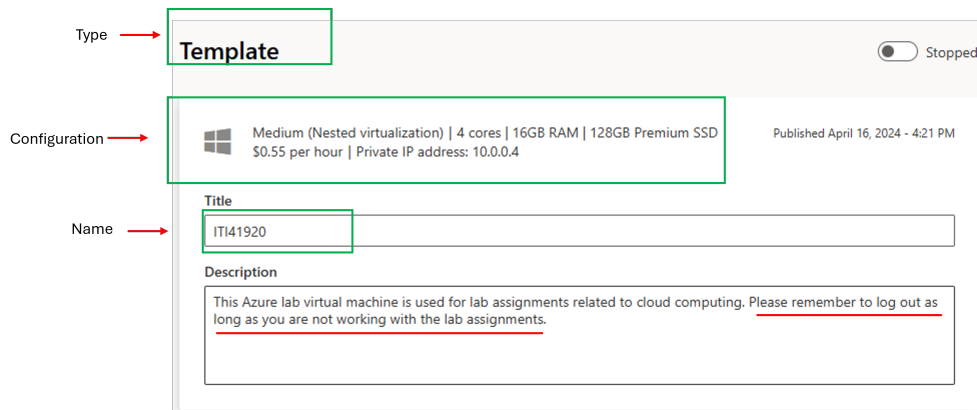


Figure 6: The template machine where you can connect to.

The Azure Labs is a cloud service provided by Microsoft. Your VM has a set of guaranteed resources with 4 CPUs, 16GB of RAM and 128GB Premium SSD. You will access the service using a Remote Desktop Connection application called remote desktop protocol (rdp). The template machine is my virtual machine - I do not have it physically. Its IP address is 10.0.0.4, falling within the range of Class A reserved for private networks, as designated by the Internet Assigned Numbers Authority (IANA). (The hardware virtualization is *nested virtualization* where you connect to my virtual machine from your virtual machine.) The IP address of your own physical PC is typically something like 192.xxx.xxx.xxx, which falls into Class C.

1. To connect to the template machine, click on the **Connect to template** symbol - Figure 7.

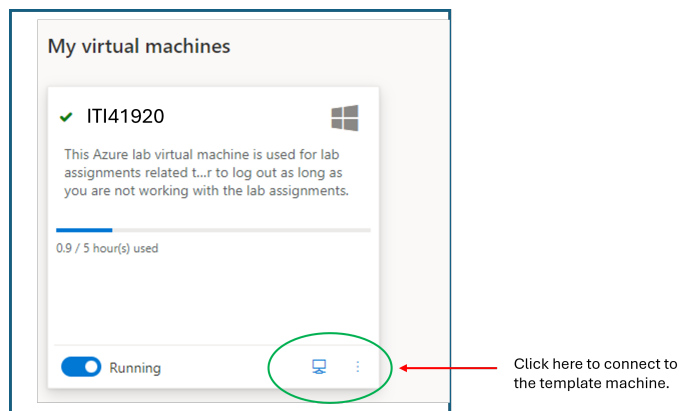


Figure 7: Connect to the Azure Labs VM (1/2).

2. Download the file **ITI41920.rdp** and open it.

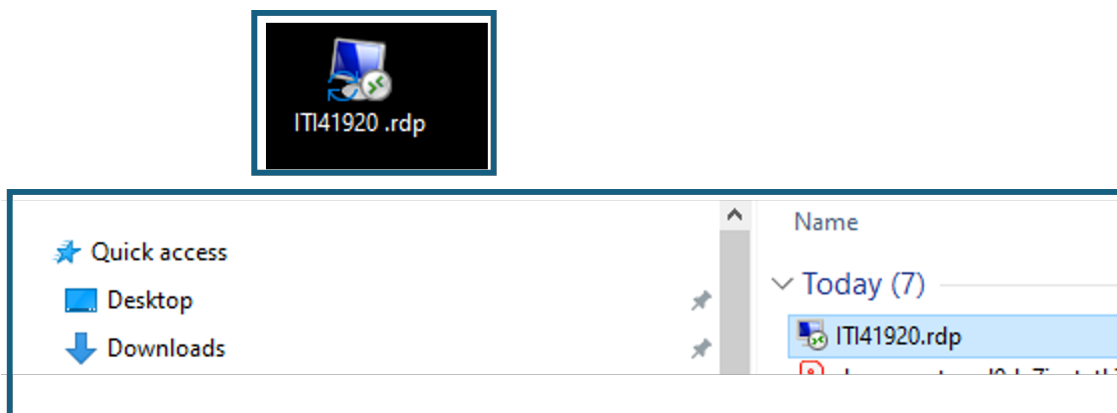


Figure 8: Connect to the Azure Labs VM (2/2).

You will see a remote desktop connection application which will connect you to the Azure Labs VM. From next time, you only need to run "ITI41920.rdp" to connect to the template machine - as long as it is ON.

3. At the login screen, login with the password as **parallel*2024** (username for this account is ITF23019) - see Figure 9 and 10.

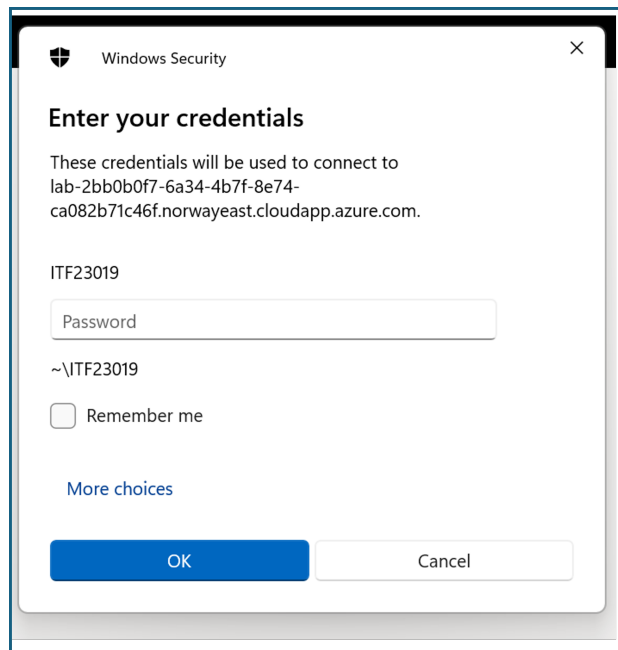


Figure 9: Remote Desktop Login (1/2).

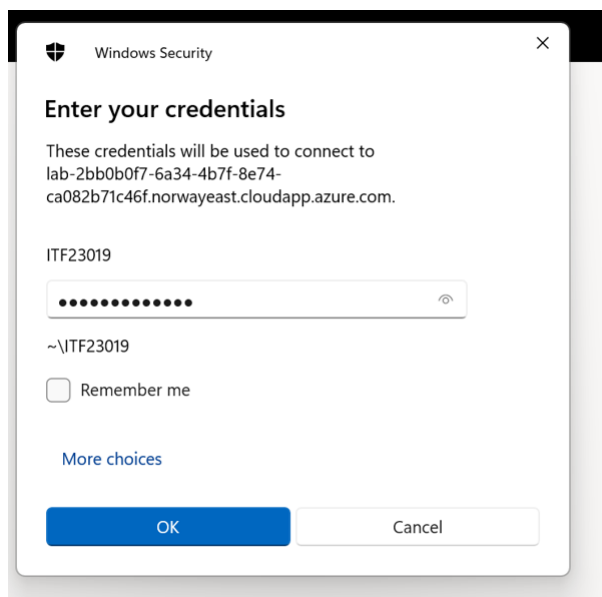


Figure 10: Remote Desktop Login (2/2).

4. You may see two security alerts. Just click on **Yes** and **Connect** buttons like the figures below to process.

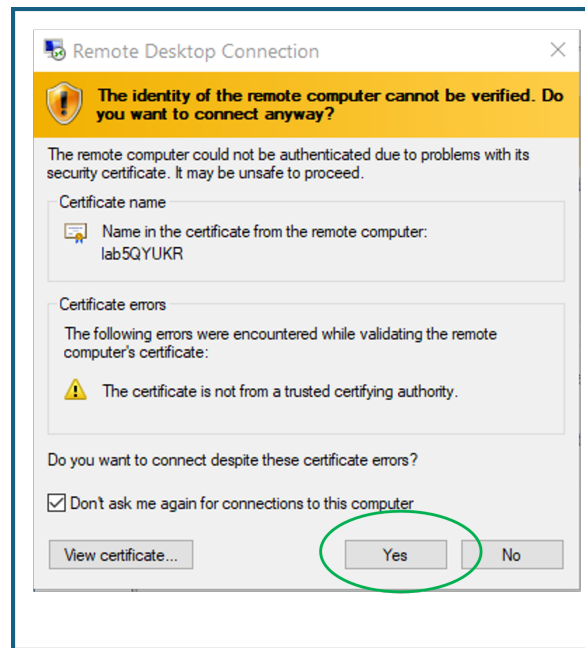


Figure 11: Security alert (1/2).

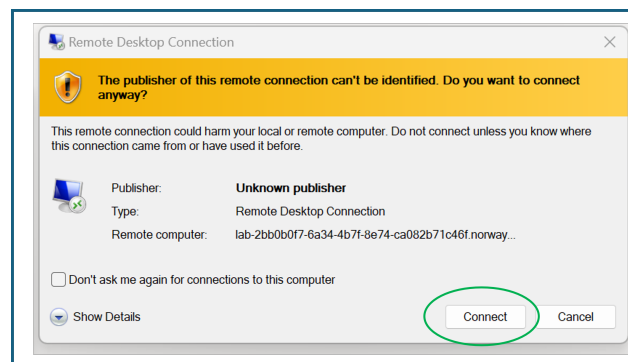


Figure 12: Security alert (2/2).

Then you will see that you are accessing the template VM - Figure 13:

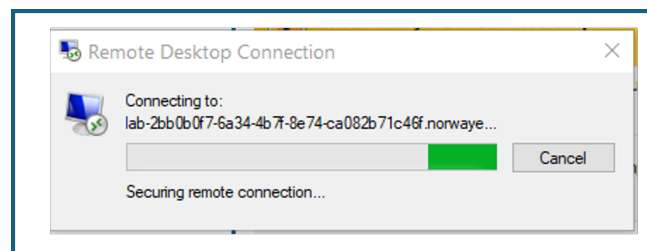


Figure 13: Logging in the template VM.

After a while, you can access the Azure Labs VM successfully - Figure 14.

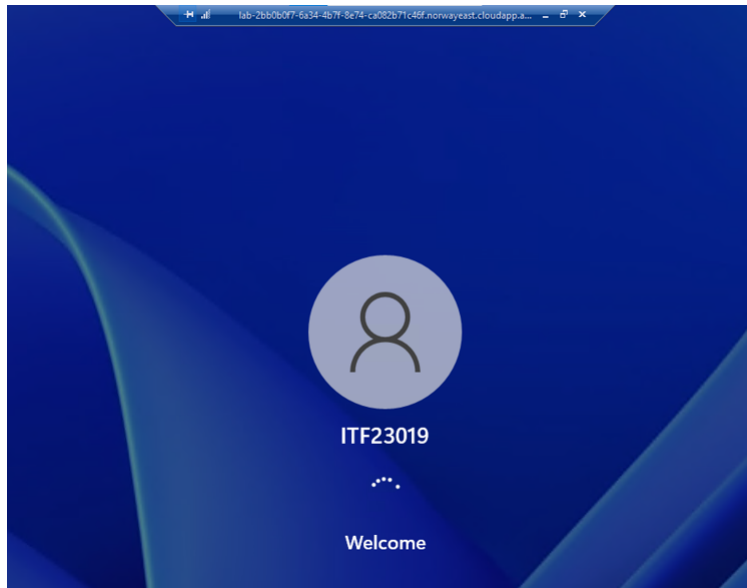


Figure 14: Accessing Azure Labs VM.

Once you have remote access to the template VM, you could see the Docker symbol in the Desktop - Figure 15.

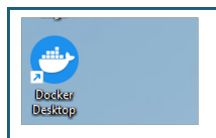


Figure 15: Docker Desktop.

2 Start the Docker Container

In this part, you will perform some practice with **Docker container**. To do so, you need to run the Docker first.

1. Click on Docker Desktop symbol in Figure 15. It can take a little too long for the Docker to start. Again, patience is important.

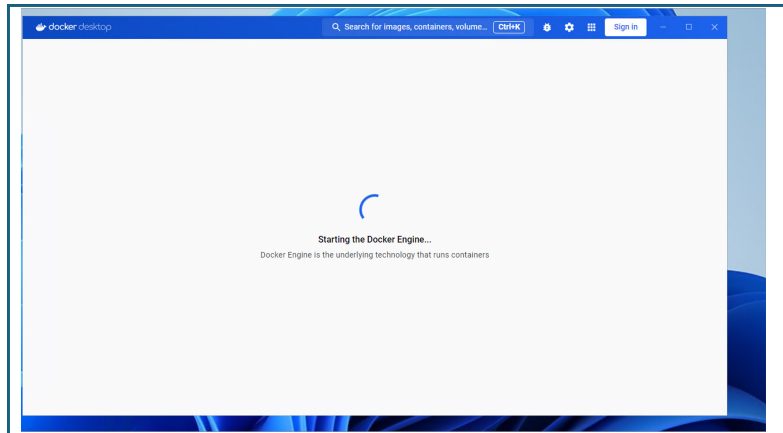


Figure 16: Starting Docker....

Once the Docker start, you can see that there are two containers: **sandbox-proxy** and **sandbox-hdp**.

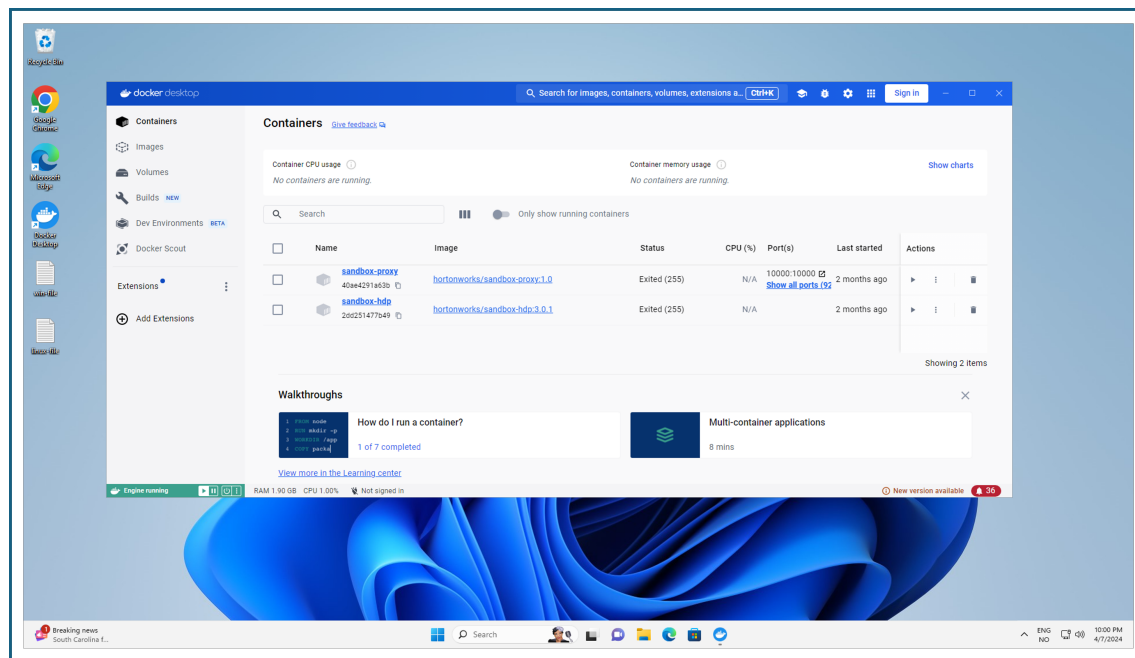


Figure 17: Docker has started.

You need to run both two containers in the Docker but start with the **sandbox-proxy** first and then **sandbox-hdp**.

3. Run **sandbox-proxy**. After the **sandbox-proxy** has started, run **sandbox-hdp**. The figure below guides you how to start a container:

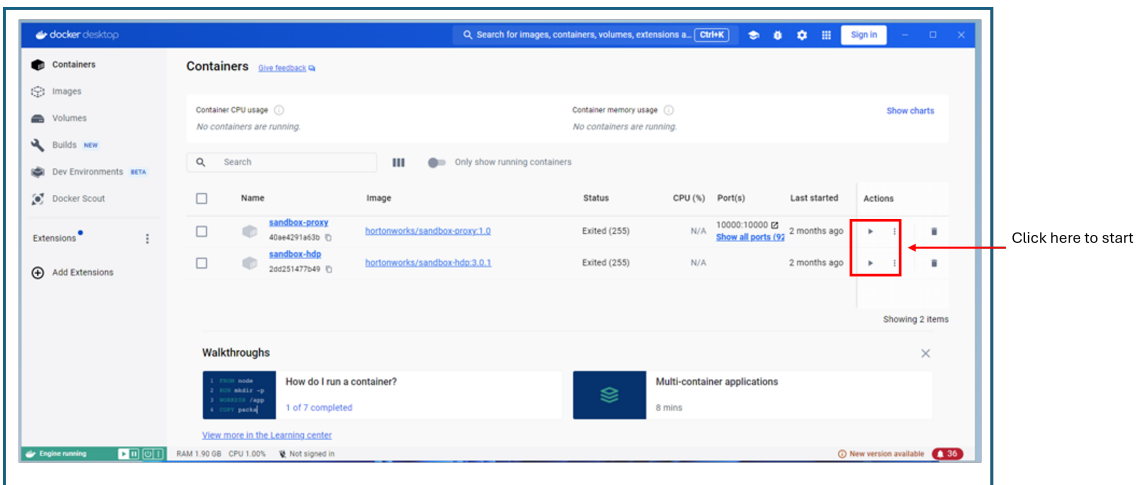


Figure 18: Starting containers in Docker.

You also can start the containers from command line. To do so, go to window prompt, and type:

```
$ docker start sandbox-proxy
$ docker start sandbox-hdp
```

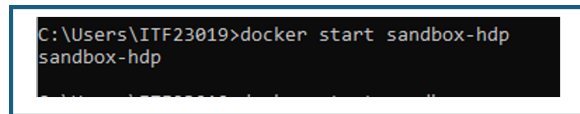


Figure 19: Alternative way to start the container.

At this step, you may experience that the `sanbox-proxy` could not start because the port number is already used by another application. It is not easy for you to manually assign another port to run this application. You can stop and try again after a while or, you can fix this problem by start the container from the command line with the following commands:

```
net stop winnat
docker start container_name
net start winnat
```

```
C:\Users\ITF23019>net stop winnat

The Windows NAT Driver service was stopped successfully.

C:\Users\ITF23019>docker start sandbox-proxy
sandbox-proxy

C:\Users\ITF23019>net start winnat

The Windows NAT Driver service was started successfully.

C:\Users\ITF23019>
```

Figure 20: Fixing the problem that the `sandbox-proxy` cannot start.

Once two containers are started successfully, you should see that the color of two containers turns green.

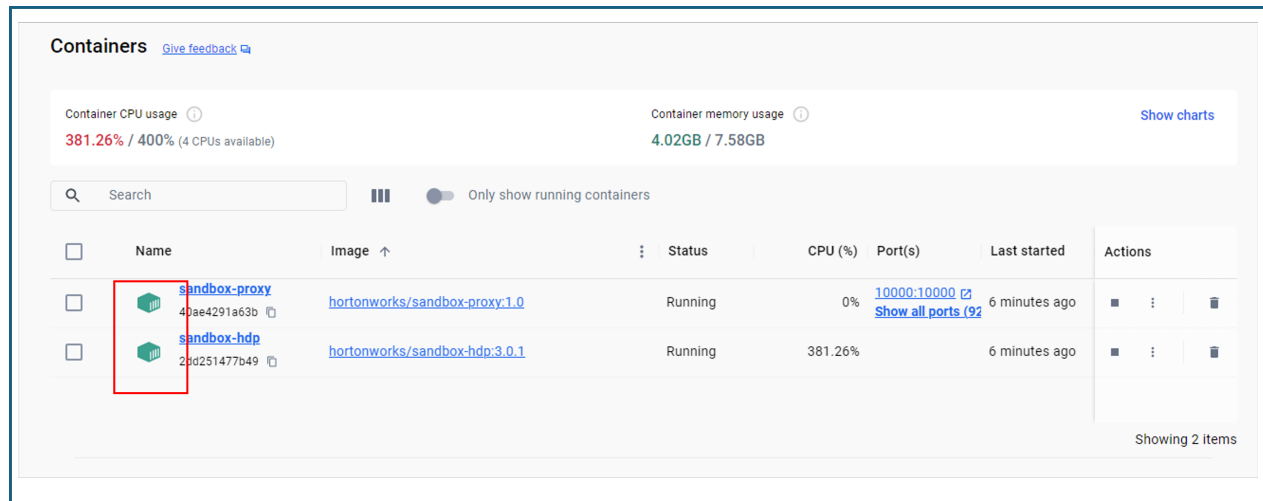


Figure 21: Two containers are running.

3 Cluster Management

Now that you can connect to the template machine and use Docker Container, let's see how you can experiment a bit.

1. Open **Shell Web Client** (i.e, **Shell-in-a-box**), it is a terminal access. You can do it simply by open a web browser.

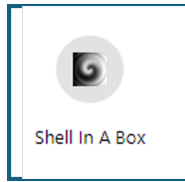


Figure 22: Shell-in-a-box.

2. Type: `localhost:8080` or `127.0.0.1:8080`.

`localhost` presents a local IP address (`127.0.0.1`) (host computer) and `8080` is the port number running the service that you are going to use. `localhost` and `8080` are IP address and port number. The service at port `8080` brings you to **Ambari** login page. Ambari is an administration tool used to monitor the status of various Hadoop components. While you may not have learned about Hadoop and Ambari, it is enough to just use them in this lab.

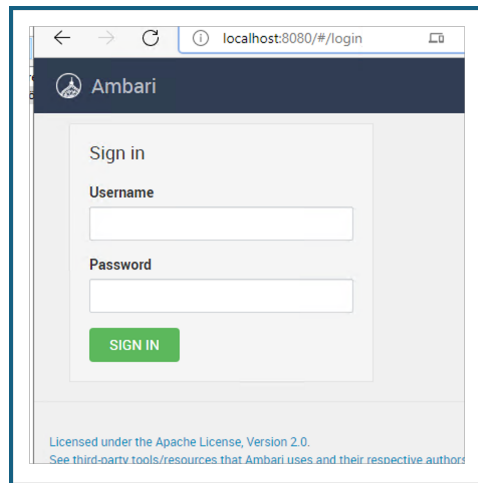


Figure 23: The Ambari Login page.

3. Login with `admin` account: `admin/ITF23019`. This brings you to the Ambari Dashboard:

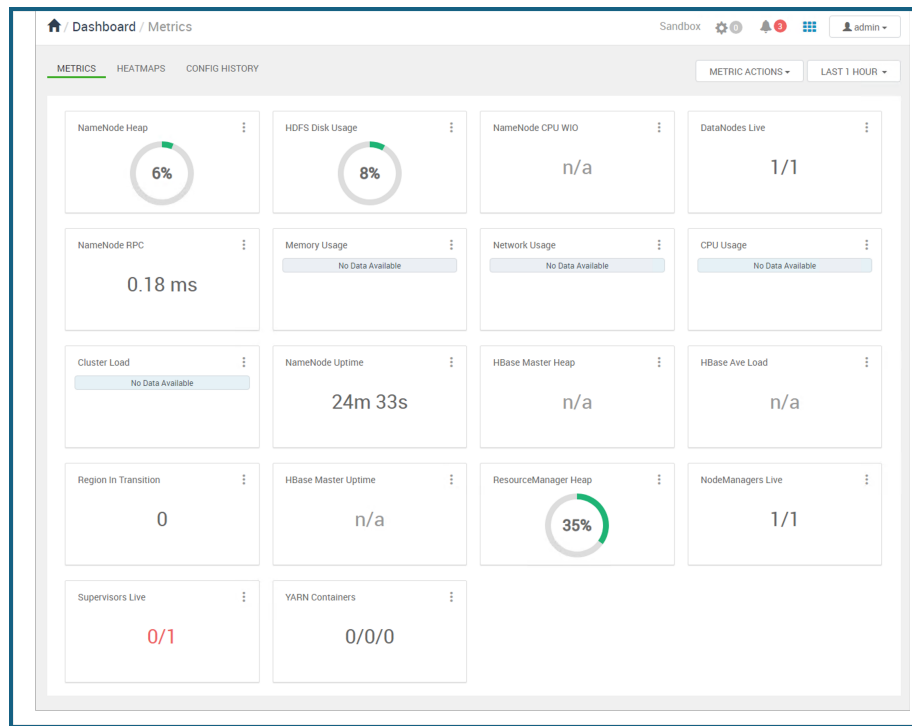


Figure 24: The Ambari Dashboard.

You can see that certain services are available:

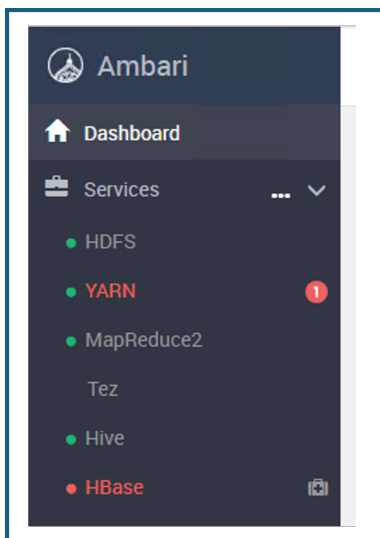


Figure 25: The Ambari Dashboard.

HDFS and MapReduce are among the services. FYI, HDFS stands for Hadoop Distributed File System. You may learn HDFS and MapReduce in next lab.

4. You can explore some information about each service, for example with HDFS:

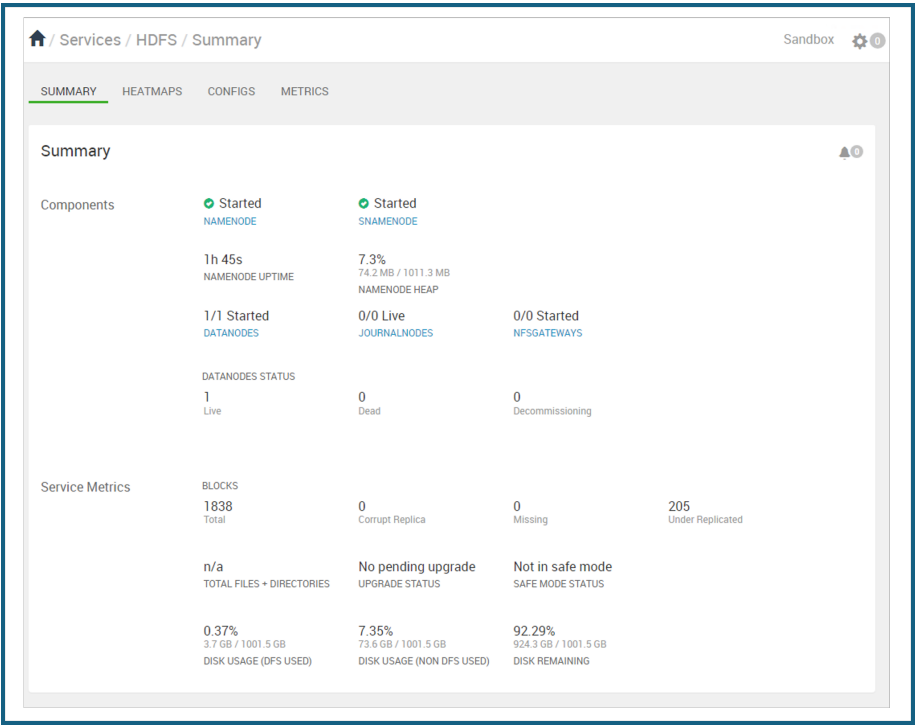


Figure 26: The HDFS service.

You can also see what actions are available with the admin account with this service:

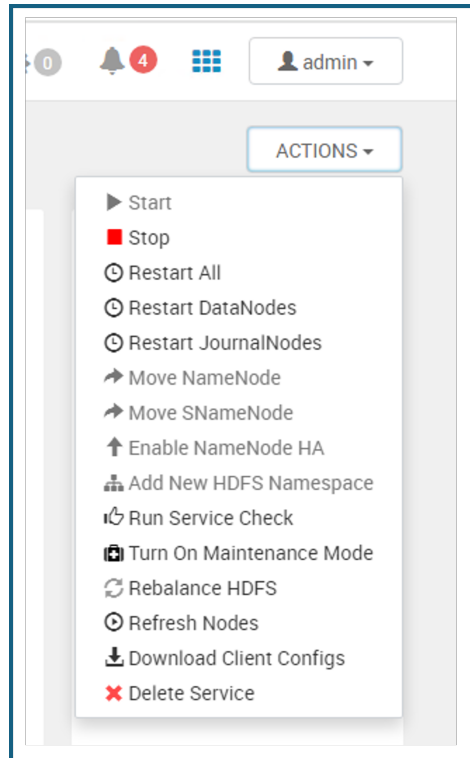


Figure 27: Action menu for HDFS service.

You can do the same with other services to explore similar information.

4 Practice with Simple File System Operations

In this section, you will practice simple file operations, including copying files to and from the Docker Container. In Docker, you can perform file operations very similar to those in the Operating System course.

4.1 Copy from Windows to Docker

1. In stead of using port 8080, let's see what service is provided in port 4200:

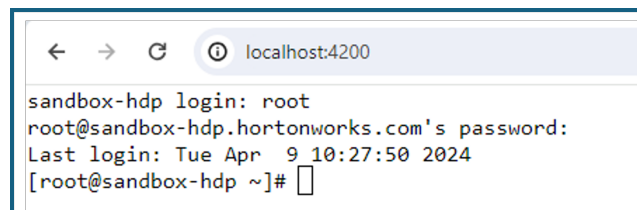
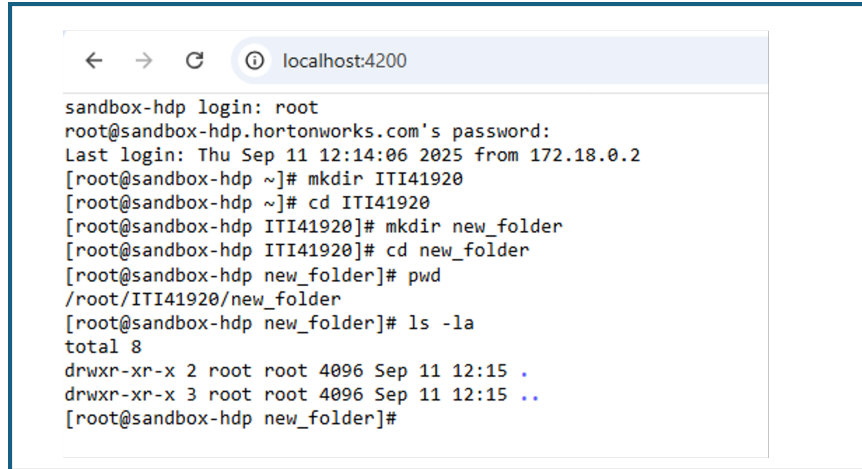


Figure 28: A Shell Web Client.

Then, login with the root account: `root/ITF23019`.

2. In **Shell Web Client**, make a directory using the `mkdir` command - Figure 28. Commands are similar with Linux commands.



```
sandbox-hdp login: root
root@sandbox-hdp.hortonworks.com's password:
Last login: Thu Sep 11 12:14:06 2025 from 172.18.0.2
[root@sandbox-hdp ~]# mkdir ITI41920
[root@sandbox-hdp ~]# cd ITI41920
[root@sandbox-hdp ITI41920]# mkdir new_folder
[root@sandbox-hdp ITI41920]# cd new_folder
[root@sandbox-hdp new_folder]# pwd
/root/ITI41920/new_folder
[root@sandbox-hdp new_folder]# ls -la
total 8
drwxr-xr-x 2 root root 4096 Sep 11 12:15 .
drwxr-xr-x 3 root root 4096 Sep 11 12:15 ..
[root@sandbox-hdp new_folder]#
```

Figure 29: Create a folder in directory: `root/ITI41920/new_folder`.

3. In **Windows**, create a file at a known location.

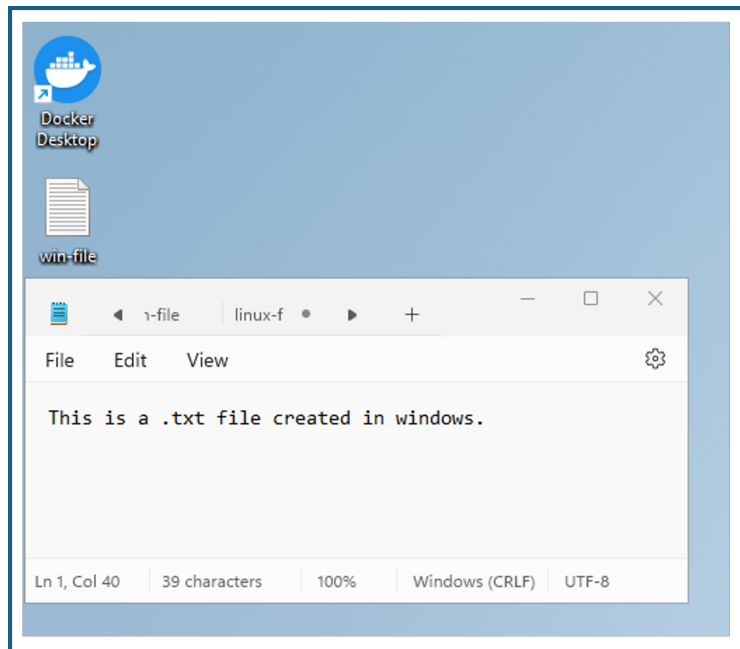


Figure 30: Create a file in Desktop Windows.

4. In **Shell Web Client** or **Powershell**, copy the new file from Windows to the file system of `sandbox-hdp` Docker container using the `docker cp` command.

```

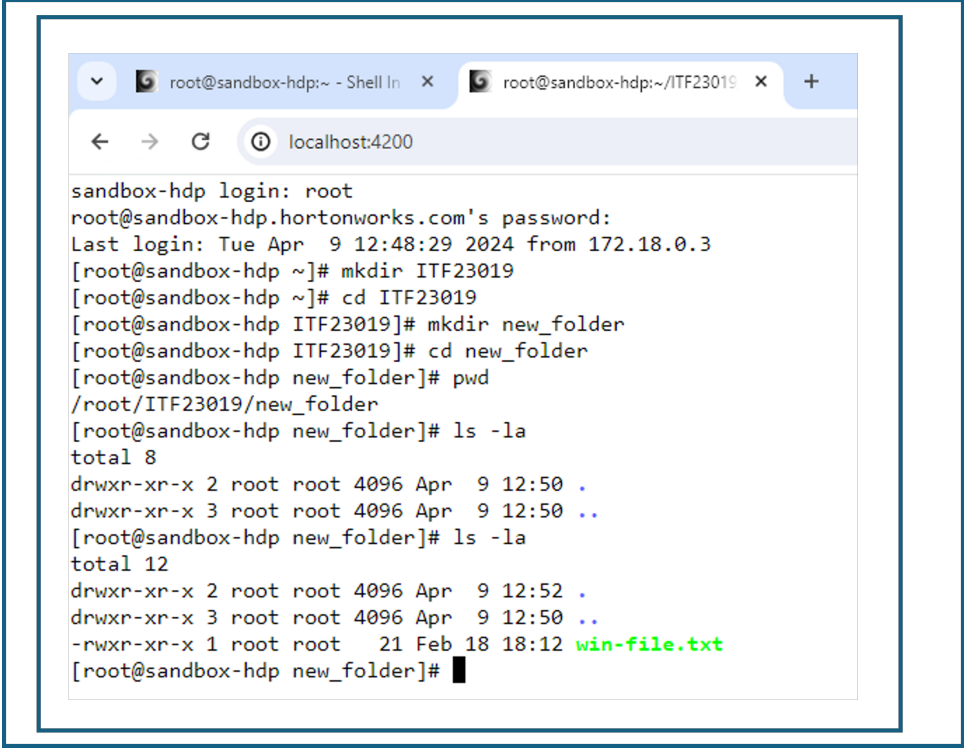
09/12/2025  08:00 AM    <DIR>          .
04/08/2025  05:19 PM    <DIR>          ..
04/16/2024  12:50 PM                2,143 Docker Desktop.lnk
09/12/2025  08:00 AM                38 win-file.txt
                2 File(s)                2,181 bytes
                2 Dir(s)  41,276,420,096 bytes free

C:\Users\ITF23019\OneDrive\Desktop>docker cp win-file.txt sandbox-hdp:root/ITF23019/new_folder
Successfully copied 2.05kB to sandbox-hdp:root/ITF23019/new_folder

```

Figure 31: Copy a file using `docker cp` command.

4. In **Shell Web Client**, verify that the file is copied successfully.



```

root@sandbox-hdp:~ - Shell In x root@sandbox-hdp:~/ITF23019 x +
localhost:4200
sandbox-hdp login: root
root@sandbox-hdp.hortonworks.com's password:
Last login: Tue Apr  9 12:48:29 2024 from 172.18.0.3
[root@sandbox-hdp ~]# mkdir ITF23019
[root@sandbox-hdp ~]# cd ITF23019
[root@sandbox-hdp ITF23019]# mkdir new_folder
[root@sandbox-hdp ITF23019]# cd new_folder
[root@sandbox-hdp new_folder]# pwd
/root/ITF23019/new_folder
[root@sandbox-hdp new_folder]# ls -la
total 8
drwxr-xr-x 2 root root 4096 Apr  9 12:50 .
drwxr-xr-x 3 root root 4096 Apr  9 12:50 ..
[root@sandbox-hdp new_folder]# ls -la
total 12
drwxr-xr-x 2 root root 4096 Apr  9 12:52 .
drwxr-xr-x 3 root root 4096 Apr  9 12:50 ..
-rwxr-xr-x 1 root root  21 Feb 18 18:12 win-file.txt
[root@sandbox-hdp new_folder]# █

```

Figure 32: Verifying copy file operation.

4.2 Copy from Docker to Windows

1. In **Shell Web Client**, create a file in Docker using `cat` command:

```
[root@sandbox-hdp ITI41920]# cd new_folder
[root@sandbox-hdp new_folder]# ls
[root@sandbox-hdp new_folder]# ls -la
total 12
drwxr-xr-x 2 root root 4096 Sep 12 08:01 .
drwxr-xr-x 3 root root 4096 Sep 11 12:15 ..
-rwxr-xr-x 1 root root  38 Sep 12 08:00 win-file.txt
[root@sandbox-hdp new_folder]# cat >linux-file.txt
This is a .txt file created in Linux.^C
[root@sandbox-hdp new_folder]# ls -la
total 12
drwxr-xr-x 2 root root 4096 Sep 12 08:18 .
drwxr-xr-x 3 root root 4096 Sep 11 12:15 ..
-rw-r--r-- 1 root root   0 Sep 12 08:18 linux-file.txt
-rwxr-xr-x 1 root root  38 Sep 12 08:00 win-file.txt
[root@sandbox-hdp new_folder]#
```

Figure 33: Create a file named `linux-file.txt`.

2. In **Powershell**, copy the newly created file into a known location in Windows:

```
C:\Users\ITF23019\OneDrive\Desktop>cd ..
C:\Users\ITF23019\OneDrive>docker cp sandbox-hdp:root/ITI41920/new_folder Desktop
Successfully copied 3.07kB to C:\Users\ITF23019\OneDrive\Desktop
C:\Users\ITF23019\OneDrive>
```

Figure 34: Copy the file named `linux-file.txt` from Docker to Windows.

3. In **Windows**, verify that the file is copied successfully.

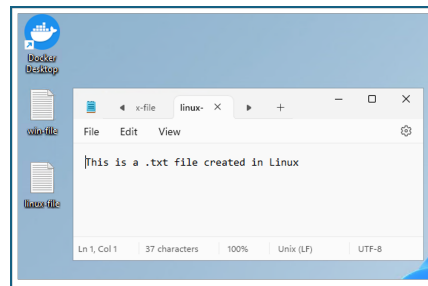


Figure 35: Verify file copy operation.

5 End of Your Session

1. Stop the two containers manually.

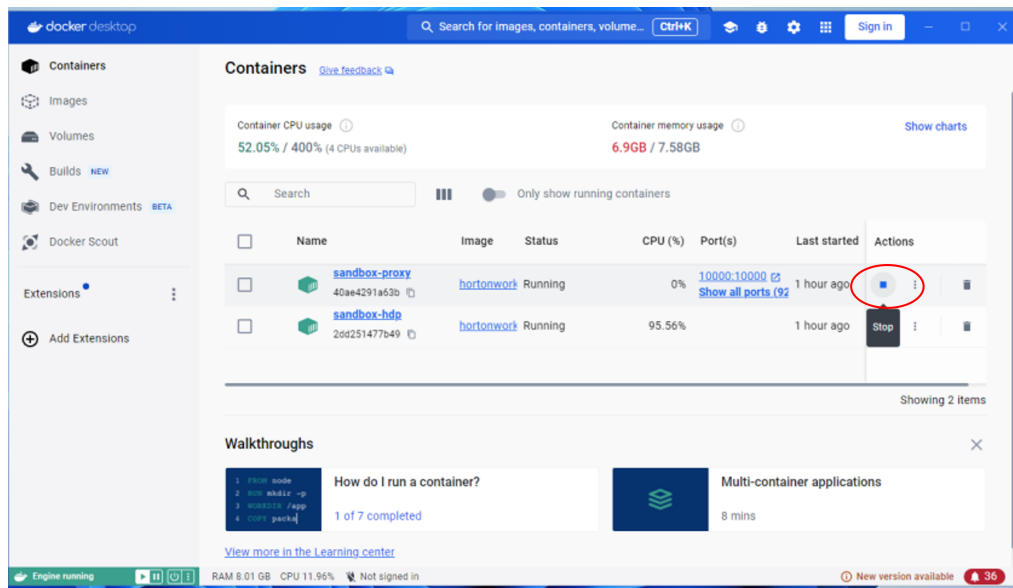


Figure 36: Stop a container.

2. Shut down Windows on the Azure labs VM from the Start button.

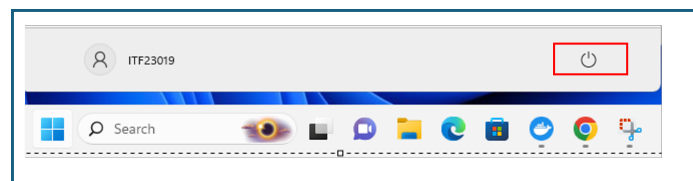


Figure 37: Shut down windows.

3. Stop the Azure Labs VM by flipping the Start/Stop switch.

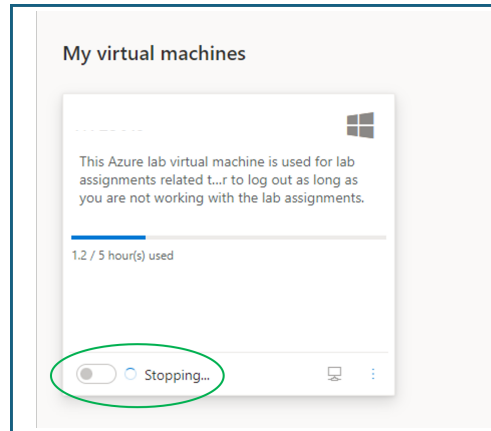
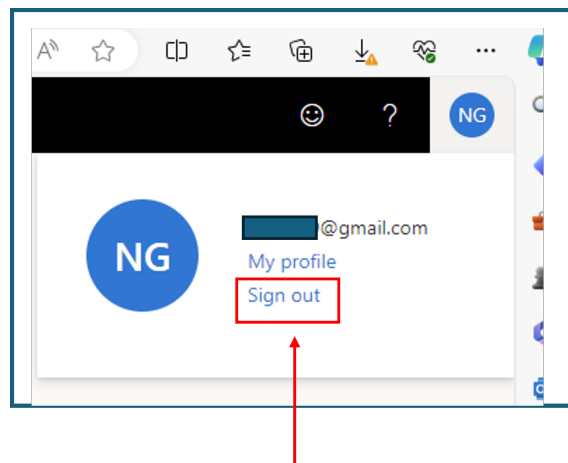


Figure 38: Shut down windows.

You may notice that service provisioning is working very well in Azure. It monitors how much time you have used for your practice. Make sure that your VM is in **Stop** state - refer to Figure 3.

4. Finally, remember to sign out.



Remember to **Sign out**.

Figure 39: The last step.

Thank you for your collaboration!ITF

6 Exercises

6.1 Exercise 1 (50 points)

Repeat subsections 4.1 and 4.2 to create `win-file.txt` and `linux-file.txt` files with different contents. Then, reproduce Figures 30 - 33, and include screenshots in your report.

6.2 Exercise 2 (50 points)

- What virtual resource do you have in this lab?
- What type of service (IaaS, PaaS, SaaS) are you using in this lab?
- Where is user interface that you used to access cloud services?
- How responsive is the Azure Labs service to your operations?

7 What To Submit

Complete the the exercises in this assignment and submit your report into Canvas.

8 Acknowledgement

1. The cost of this lab is covered by our university college.
2. Some parts of this lab are sourced (with agreement) Hogskolan Danarna.